

SPARK: Matrix-Gate Folding

A multilinear polynomial commitment IOP

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Abstract

We construct a family of linear codes for multilinear polynomials from GL_2 gate matrices attached to the nodes of a binary tree, a folding operator that is exactly the gate rule evaluated at a random point, and a commitment scheme in which the folding challenges are the opening point. The construction uses no structured evaluation domain: gates are arbitrary elements of $\text{GL}_2(\mathbb{F})$ with nonzero second row, and the encoder uses only operations of the form $u + tv$. We prove the algebraic identities; three distance theorems for randomly sampled per-node gates (a one-layer bound, a rank-saturation bound, and a bound with a checked setup); that the generic code is MDS; a zero decomposition and an obstruction showing what any sharper distance bound must pay; a proximity lemma and an agreement lemma for the folding operator; and soundness of the commitment with a per-query detection rate of $\min(1 - (1 - \Delta)^{1/3}, \Delta/2)$, with no loss in the number of variables. The construction works over any field large enough for the distance bounds; gates may be sampled in an extension field. A Rust implementation accompanies this document.

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1 Notation

- $\mathbb{F}$ is a finite field with $q = |\mathbb{F}|$ (any characteristic, including 2); $\mathbb{K} \supseteq \mathbb{F}$ is an extension field used for verifier challenges.
- $\text{ML}(x_j, \dots, x_n)$ denotes multilinear polynomials in the listed variables; ML_i denotes multilinear polynomials in i variables.
- $k \geq 0$ is the *redundancy parameter*. For $0 \leq i \leq n$ put

$$\Omega_i = \{0, 1\}^k \times \{0, 1\}^i, \quad L_i = |\Omega_i| = 2^{k+i},$$

and identify $\Omega_i = \Omega_{i-1} \times \{0, 1\}$, writing $\omega = (\omega', b)$.

- A position $\omega \in \Omega_n$ is written $\omega = (\beta, \theta_n, \theta_{n-1}, \dots, \theta_1)$ with $\beta \in \{0, 1\}^k$, and its prefixes are $\omega^{(j)} = (\beta, \theta_n, \dots, \theta_{j+1}) \in \Omega_{n-j}$ for $0 \leq j \leq n$.
- For $w, w' \in \mathbb{K}^{\Omega_i}$: $d_{\text{pos}}(w, w') = L_i^{-1} |\{\omega : w(\omega) \neq w'(\omega)\}|$, and for $i \geq 1$ the *block distance*

$$d_{\text{blk}}(w, w') = L_{i-1}^{-1} |\{\omega' \in \Omega_{i-1} : (w(\omega', 0), w(\omega', 1)) \neq (w'(\omega', 0), w'(\omega', 1))\}|.$$

Always $d_{\text{pos}} \leq d_{\text{blk}} \leq 2d_{\text{pos}}$. Distances to a set are minima. For a linear code C , $\Delta(C)$ is its minimum relative distance.

2 Gates

Definition 2.1 (Gate). A *gate* is a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2(\mathbb{F})$ with $c \neq 0$, $d \neq 0$. Write $\delta = ad - bc$ and $\mathcal{G}^* = \{A \in \text{GL}_2(\mathbb{F}) : c \neq 0, d \neq 0\}$. Its *gate weights* are $W_L(x) = a + cx$ and $W_R(x) = b + dx$, and its *structural points* are

$$t^0 = -\frac{a}{c}, \quad t^1 = -\frac{b}{d}.$$

Lemma 2.2 (Gates are point pairs with scales). *The map $(c, d, t^0, t^1) \mapsto \begin{pmatrix} -t^0 c & -t^1 d \\ c & d \end{pmatrix}$ is a bijection from $(\mathbb{F}^\times)^2 \times \{(t^0, t^1) \in \mathbb{F}^2 : t^0 \neq t^1\}$ onto $\mathcal{G}^*$, and $\delta = cd(t^1 - t^0)$. In particular, if A is uniform on $\mathcal{G}^*$ then (t^0, t^1) is uniform on pairs of distinct elements.*

Proof. With $a = -t^0 c$, $b = -t^1 d$ one gets $ad - bc = -t^0 cd + t^1 cd = cd(t^1 - t^0)$, nonzero iff $t^0 \neq t^1$. The inverse map is $A \mapsto (c, d, -a/c, -b/d)$. The fibre over each (t^0, t^1) has size $(q-1)^2$. $\square$

Lemma 2.3 (Decomposition and collapse). *Let $A \in \mathcal{G}^*$ and let $F = u + xv$ with u, v independent of x . There are unique F_L, F_R independent of x with*

$$F = W_L(x) F_L + W_R(x) F_R, \quad F_L = \delta^{-1}(du - bv), \quad F_R = \delta^{-1}(-cu + av).$$

Moreover

$$F|_{x=t^0} = \gamma^0 F_R, \quad \gamma^0 = -\frac{\delta}{c}, \quad F|_{x=t^1} = \gamma^1 F_L, \quad \gamma^1 = +\frac{\delta}{d}.$$

Proof. $W_L F_L + W_R F_R = (aF_L + bF_R) + x(cF_L + dF_R)$, so the identity is $A_{(F_R)}^{(F_L)} = \binom{u}{v}$, uniquely solvable since $\delta \neq 0$. At $x = t^0$: $W_L(t^0) = 0$ and $W_R(t^0) = b - ad/c = -\delta/c$. At $x = t^1$: $W_R(t^1) = 0$ and $W_L(t^1) = a - bc/d = \delta/d$. $\square$

Lemma 2.4 (Gate rule at an arbitrary point). *In the setting of Lemma 2.3, put $w^0 = F|_{x=t^0}$ and $w^1 = F|_{x=t^1}$. For every z ,*

$$F|_{x=z} = W_L(z) \frac{d}{\delta} w^1 - W_R(z) \frac{c}{\delta} w^0 = w^0 + \frac{z - t^0}{t^1 - t^0} (w^1 - w^0).$$

Proof. By Lemma 2.3, $F_L = (d/\delta)w^1$ and $F_R = -(c/\delta)w^0$; substitute into $F = W_L F_L + W_R F_R$ at $x = z$. The second expression is the unique affine function of z taking the values w^0, w^1 at t^0, t^1 . $\square$

Lemma 2.5 (One-layer sum rule). *If $F = W_L(x)F_L + W_R(x)F_R$ then $\sum_{x \in \{0,1\}} F = (2a + c)F_L + (2b + d)F_R$.*

Proof. $\sum_{x \in \{0,1\}} (a + cx) = 2a + c$ and $\sum_{x \in \{0,1\}} (b + dx) = 2b + d$. $\square$

3 The gate code

Definition 3.1 (Gate family). A *gate family* is a collection $\mathcal{A} = (A_j(\omega'))_{1 \leq j \leq n, \omega' \in \Omega_{n-j}}$ of gates, one per variable x_j and per node $\omega' \in \Omega_{n-j}$. Write $t_j^b(\omega')$ for the structural points of $A_j(\omega')$. The family is *per-layer* if $A_j(\omega')$ does not depend on ω' , and *per-node* otherwise.

Definition 3.2 (Encoding). For $0 \leq i \leq n$ let C_i encode $\text{ML}(x_{n-i+1}, \dots, x_n)$ as follows.

$$\text{Enc}_0(c)(\beta) = c \quad (\beta \in \{0, 1\}^k),$$

and for $i \geq 1$, writing $j = n - i + 1$ and $F = u + x_j v$ with $u, v \in \text{ML}(x_{j+1}, \dots, x_n)$,

$$\text{Enc}_i(F)(\omega', b) = \text{Enc}_{i-1}(u)(\omega') + t_j^b(\omega') \text{Enc}_{i-1}(v)(\omega'), \quad \omega' \in \Omega_{i-1}, b \in \{0, 1\}.$$

Put $C_i = \text{Enc}_i(\text{ML}(x_{n-i+1}, \dots, x_n)) \subseteq \mathbb{F}^{\Omega_i}$.

Proposition 3.3 (Closed form). *For $F \in \text{ML}(x_1, \dots, x_n)$ and $\omega = (\beta, \theta_n, \dots, \theta_1) \in \Omega_n$,*

$$\text{Enc}_n(F)(\omega) = F(r(\omega)), \quad r_j(\omega) = t_j^{\theta_j}(\omega^{(j)}).$$

The coordinate $r_j(\omega)$ depends only on $(\beta, \theta_n, \dots, \theta_j)$. Hence two positions that differ only in θ_j have points differing only in the coordinate x_j .

Proof. Induction on n . For $F = u + x_1 v$, $\text{Enc}_n(F)(\omega^{(1)}, \theta_1) = u(r_{\geq 2}) + t_1^{\theta_1}(\omega^{(1)}) v(r_{\geq 2}) = F(r(\omega))$, where $r_{\geq 2}$ is the point attached to $\omega^{(1)}$ by the induction hypothesis. $\square$

Remark 3.4 (Relation to the bottom table). By Lemma 2.3, the bottom table T_0 of the original symbolic folding protocol satisfies $T_0[\theta] = \Gamma(\theta)^{-1} F(r(\theta))$ with a public nonzero $\Gamma(\theta)$. So T_0 is $\text{Enc}_n(F)$ for $k = 0$ and a per-layer family, up to a public diagonal scaling, which preserves all distances. Propositions 3.7 and 3.8 explain why that table alone cannot support spot-checks.

Proposition 3.5 (Encoding cost). *$\text{Enc}_n(F)$ is computed with at most $(n+1)2^{n+k}$ multiplications and as many additions in $\mathbb{F}$.*

Proof. With $T(i)$ the cost for i variables, $T(i) = 2T(i-1) + L_i$ and $T(0) = L_0 = 2^k$, so $T(n) = (n+1)2^{n+k}$. $\square$

Lemma 3.6 (Extension of scalars). *For a linear code $C \subseteq \mathbb{F}^N$, the code $C \otimes \mathbb{K} \subseteq \mathbb{K}^N$ has $\Delta(C \otimes \mathbb{K}) = \Delta(C)$.*

Proof. Fix an $\mathbb{F}$ -basis (e_l) of $\mathbb{K}$. A word of $C \otimes \mathbb{K}$ is $\sum_l e_l c_l$ with $c_l \in C$. At a position p , $\sum_l e_l c_l(p) = 0$ iff all $c_l(p) = 0$, so the support is $\bigcup_l \text{supp } c_l$. $\square$

Proposition 3.7 (Per-layer families have distance 2^{-n}). *If $\mathcal{A}$ is per-layer, then $\Delta(C_n) \leq 2^{-n}$.*

Proof. All points $r(\omega)$ lie in the grid $\prod_j \{t_j^0, t_j^1\}$, each grid point appearing 2^k times. $F = \prod_{j=1}^n (x_j - t_j^0)$ is nonzero exactly at the grid point $(t_1^1, \dots, t_n^1)$, so $\text{wt}(\text{Enc}_n(F)) = 2^k$, of relative weight 2^{-n} . $\square$

Proposition 3.8 (No redundancy, no testing). *If $k = 0$ and $\Delta(C_n) > 0$, then $C_n = \mathbb{F}^{\Omega_n}$. Every table is then a codeword, and $\Delta(C_n) = 2^{-n}$.*

Proof. Enc_n is injective and $\dim C_n = 2^n = L_n$. $\square$

3.1 Distance of per-node random families

Lemma 3.9 (One-layer distance). *Let $C \subseteq \mathbb{F}^N$ be linear with $\dim C = D$ and $\Delta(C) \geq \Delta$. Let $(t_p^0, t_p^1)_{p \leq N}$ be independent, each uniform on pairs of distinct elements of $\mathbb{F}$, and*

$$C' = \{(c_u + t^0 \circ c_v, c_u + t^1 \circ c_v) : c_u, c_v \in C\} \subseteq \mathbb{F}^{2N},$$

with coordinates paired as $(p, 0), (p, 1)$. For every integer $s \geq 0$,

$$\Pr\left[\Delta(C') < \Delta - \frac{s}{2N} \text{ or } \dim C' < 2D\right] \leq q^{2D} \binom{N}{s} \left(\frac{2}{q}\right)^s.$$

Proof. Fix $(c_u, c_v) \neq (0, 0)$ and let w be the corresponding word of C' .

If $c_v = 0$, then $w = (c_u, c_u)$ and $\text{wt}(w) = 2 \text{wt}(c_u) \geq 2\Delta N$.

If $c_v \neq 0$, consider $p \in \text{supp } c_v$. The pair $(c_u(p) + t_p^0 c_v(p), c_u(p) + t_p^1 c_v(p))$ has at most one zero, since $t_p^0 \neq t_p^1$. It has a zero iff $\alpha_p := -c_u(p)/c_v(p) \in \{t_p^0, t_p^1\}$, an event of probability exactly $2/q$, independently over p . Let Z be the set of such p . Then $\text{wt}(w) \geq 2 \text{wt}(c_v) - |Z| \geq 2\Delta N - |Z|$, and $\Pr[|Z| \geq s] \leq \binom{N}{s} (2/q)^s$.

A union bound over the q^{2D} pairs gives: outside the stated event every nonzero pair has relative weight $\geq \Delta - s/(2N) > 0$ (when this is positive), which also gives injectivity, hence $\dim C' = 2D$. $\square$

Theorem 3.10 (Distance of the gate code). *Let $\mathcal{A}$ be a per-node family with independent gates, each uniform on $\mathcal{G}^*$. For integers $s_1, \dots, s_n \geq 0$, with probability at least*

$$1 - \sum_{i=1}^n q^{2^i} \binom{L_{i-1}}{s_i} \left(\frac{2}{q}\right)^{s_i},$$

every C_i ($0 \leq i \leq n$) has dimension 2^i and

$$\Delta(C_i) \geq \Delta^* := 1 - \sum_{l=1}^n \frac{s_l}{2L_{l-1}}.$$

Proof. C_0 is the repetition code, with $\Delta(C_0) = 1$ and dimension 1. Definition 3.2 builds C_i from C_{i-1} exactly as C' from C in Lemma 3.9, with $N = L_{i-1}$, $D = 2^{i-1}$ and fresh structural points (uniform distinct pairs by Lemma 2.2). Apply Lemma 3.9 at each level and take a union bound. $\square$

Remark 3.11 (Parameters). Using $\binom{N}{s} \leq (eN/s)^s$, Table 1 lists the smallest certified Δ^* with setup failure probability $\leq 2^{-128}$. Sample gates in the base field $\mathbb{F}$. By Lemma 3.6, the distance carries over to $\mathbb{K}$.

$\log_2 q$	n	k	certified Δ^*
31	16	5	0.212
31	16	6	0.589
31	20	6	0.508
31	24	6	0.427
62	20	6	0.604
124	20	5	0.287
124	20	6	0.640

Table 1: Certified distance (Theorem 3.10), setup failure $\leq 2^{-128}$. For $k \leq 4$ the certificate is vacuous at these n .

q	n	k	length	per-node min	per-node mean	per-layer
13	2	3	32	0.563	0.625	0.250
61	2	3	32	0.781	0.797	—
7	3	2	32	0.250	0.339	0.125
7	3	3	64	0.375	0.412	0.125
3	4	2	64	0.078	0.096	0.0625

Table 2: Exhaustive minimum relative distance over random families (6 samples, 4 for $q = 61$). Per-layer families sit exactly at 2^{-n} (Proposition 3.7). Per-node distance grows with q , as Lemma 3.9 predicts ($2/q$ collision rate).

3.2 Distance by rank saturation

Theorem 3.10 loses about 2^{-k} per level. The argument below loses only about $H_2(2^{-k})/\log_2 q$ per level, so large fields allow small redundancy.

For a set X of positions of C_i write $V_X = \{F : \text{Enc}_i(F)|_X = 0\}$ and $d_i(X) = \dim V_X$.

Lemma 3.12 (Augmentation). *Let $C = \text{Enc}(\text{ML})$ be injective and let $Z \geq 1$. If $V_S = 0$ for every S with $|S| \geq Z$, then $d(X) \leq \max(0, Z - |X|)$ for every X .*

Proof. Suppose $d(X) = d \geq 1$. Since Enc is injective, some element of V_X is nonzero at some position ω , and $d(X \cup \{\omega\}) = d - 1$. Repeating $d - 1$ times gives Y with $|Y| = d - 1$ and $V_{X \cup Y} \neq 0$. Hence $|X| + d - 1 < Z$. $\square$

Lemma 3.13 (One level). *Fix the gates of C_{i-1} , and assume $d_{i-1}(X) \leq \max(0, Z' - |X|)$ for all X . Let $S \subseteq \Omega_i$ with $|S| \geq 2Z'$. Over the level- i gates,*

$$\Pr[V_S \neq 0] \leq \frac{(1 + 1/q)^{L_{i-1}}}{q - 1} q^{2Z' - |S|}.$$

Proof. Let T_2 be the blocks with both entries in S and T_1 the blocks with exactly one, say $(\omega', b_{\omega'})$; then $|S| = 2|T_2| + |T_1|$. Write $F = u + xv$ and $U = V_{T_2}$ (level $i - 1$), and put $\hat{u} = \text{Enc}_{i-1}(u)$, $\hat{v} = \text{Enc}_{i-1}(v)$.

Constraints. $F \in V_S$ iff:

- $u, v \in U$: on a double block, $\hat{u} + t^0 \hat{v} = \hat{u} + t^1 \hat{v} = 0$ with $t^0 \neq t^1$ forces $\hat{u} = \hat{v} = 0$;
- for every $\omega' \in T_1$: $\hat{u}(\omega') + t \hat{v}(\omega') = 0$, where $t = t^{b_{\omega'}}(\omega')$ is uniform on $\mathbb{F}$ and independent across blocks.

For fixed (u, v) , the constraint at $\omega' \in T_1$ holds with probability $1/q$ if $\hat{v}(\omega') \neq 0$, and with probability $[\hat{u}(\omega') = 0]$ if $\hat{v}(\omega') = 0$.

Counting. V_S is a subspace, so $\Pr[V_S \neq 0] \leq \mathbb{E}[\#(V_S \setminus 0)]/(q - 1)$.

- Pairs with $v = 0$ contribute $q^{d_{i-1}(T_2 \cup T_1)} - 1$. This is 0, because $|T_2| + |T_1| \geq |S|/2 \geq Z'$.
- For $v \neq 0$, let $N(v) = \{\omega' \in T_1 : \hat{v}(\omega') = 0\}$. Summing over u gives the weight $q^{|N(v)| - |T_1|} q^{d_{i-1}(T_2 \cup N(v))}$.
- Group the $v \neq 0$ by $N = N(v)$. They lie in $V_{T_2 \cup N} \setminus 0$, which is nonempty only if $d_{i-1}(T_2 \cup N) \geq 1$, i.e. $|T_2| + |N| < Z'$. Then $d_{i-1}(T_2 \cup N) \leq Z' - |T_2| - |N|$.

Hence

$$\mathbb{E}[\#(V_S \setminus 0)] \leq \sum_{N \subseteq T_1} q^{|N| - |T_1| + 2(Z' - |T_2| - |N|)} = q^{2Z' - |S|} \sum_{N \subseteq T_1} q^{-|N|} = q^{2Z' - |S|} (1 + 1/q)^{|T_1|}. \quad \square$$

Theorem 3.14 (Distance by rank saturation). *Let the gates be independent and uniform on $\mathcal{G}^*(\mathbb{F})$. Choose integers $g_1, \dots, g_n \geq 0$, put $Z_0 = 1$ and $Z_i = 2Z_{i-1} + g_i$, and assume $Z_i \leq L_i$. With probability at least*

$$1 - \sum_{i=1}^n \binom{L_i}{Z_i} \frac{(1 + 1/q)^{L_{i-1}}}{q - 1} q^{-g_i},$$

every nonzero codeword of every C_i has fewer than Z_i zeros. In particular $\Delta(C_i) \geq 1 - (Z_i - 1)/L_i$, and $\min_i \Delta(C_i) \geq 1 - (Z_n - 1)/L_n$.

Proof. Induction on i .

- *Base.* C_0 is the repetition code, so $V_S = 0$ for every nonempty S , i.e. $Z_0 = 1$.
- *Step.* Assume the statement for C_{i-1} . Then Enc_{i-1} is injective (because $Z_{i-1} \leq L_{i-1}$), and Lemma 3.12 gives $d_{i-1}(X) \leq \max(0, Z_{i-1} - |X|)$ for all X .
- *Union bound.* By Lemma 3.13 with $Z' = Z_{i-1}$, each S with $|S| = Z_i$ satisfies $V_S \neq 0$ with probability at most $\frac{(1+1/q)^{L_{i-1}}}{q-1} q^{-g_i}$. A union bound over the $\binom{L_i}{Z_i}$ such sets, together with monotonicity in S , gives $V_S = 0$ for all $|S| \geq Z_i$.

The excess $(Z_i - D_i)/L_i$ is nondecreasing in i and $1/L_i$ decreases, so C_n attains the minimum. $\square$

gate field	n	copies	Δ (Thm. 3.14)	queries	Δ (Thm. 3.10)
Goldilocks (2^{64})	20	16	0.579	327	< 0
Goldilocks (2^{64})	20	32	0.729	218	0.221
Goldilocks (2^{64})	20	64	0.811	183	0.605
Goldilocks ³ (2^{192})	20	4	0.675	253	< 0
Goldilocks ³ (2^{192})	20	8	0.812	183	< 0
M31 (2^{31})	20	64	0.417	523	0.508

Table 3: Certified distance with setup failure $\leq 2^{-128}$, and the resulting queries for 2^{-128} soundness (Theorem 6.4, $\delta = 0.95 \delta_{\max}$). Both theorems are valid, so the larger value applies.

Remark 3.15 (Size of the loss). Writing $Z_i = D_i + e_i$ gives $e_i = 2e_{i-1} + g_i$, so $e_i/L_i = e_{i-1}/L_{i-1} + g_i/L_i$. The union bound needs $g_i \log_2 q \gtrsim L_i H_2(Z_i/L_i) + \lambda$. So the relative loss per level is about $H_2(2^{-k})/\log_2 q$, plus $\lambda/(L_i \log_2 q)$ on the first few levels. Gates may be sampled in an extension $\mathbb{F}' \supseteq \mathbb{F}$, at the cost of storing E_0 in $\mathbb{F}'$.

Theorem 3.16 (The generic code is MDS). *Regard the structural points $t_i^b(\omega')$ as independent indeterminates, and work over the field of rational functions in them. Then every set of at most D_i coordinate functionals of C_i is linearly independent; equivalently, the coordinate matroid of C_i is uniform of rank D_i , and the generic code C_n is MDS.*

Proof. Induction on i . For $i = 0$ every coordinate functional is $c \mapsto c \neq 0$.

Step. Identify ML_i with ML_{i-1}^2 via $F = u + xv$. Position (ω', b) has functional $(\ell_{\omega'}, t^b \ell_{\omega'})$, where $\ell_{\omega'}$ is the coordinate functional of C_{i-1} at ω' . Let S be a set of positions with $|S| \leq 2D'$, $D' = D_{i-1}$. Let A be its double blocks and B its single blocks, with $a = |A|$ and $b = |B|$, so $2a + b \leq 2D'$.

Double blocks. Since $t^1 - t^0 \neq 0$, the two rows of a double block span $(\ell_{\omega'}, 0)$ and $(0, \ell_{\omega'})$. By induction the $\ell_{\omega'}$, $\omega' \in A$, are independent, because $a \leq D'$. So the double blocks contribute rank $2a$ and span $U \oplus U$, with $U = \text{span}\{\ell_{\omega'} : \omega' \in A\}$.

Quotient. Pass to $W = \text{ML}_{i-1}^*/U$, of dimension $m = D' - a$, and write $\bar{\ell}_{\omega'}$ for the images. By induction, any at most m of the $\bar{\ell}_{\omega'}$, $\omega' \in B$, are independent: if $|T| \leq m$ then $|A \cup T| \leq D'$.

Single blocks. Since $b \leq 2m$, split $B = B_0 \sqcup B_1$ with $|B_0|, |B_1| \leq m$. Choose coordinates J_0 and J_1 of W with $|J_r| = |B_r|$ such that the minors $\det M_{B_0}$ and $\det M_{B_1}$ of $\{\bar{\ell}_{\omega'}\}$ are nonzero.

Consider the $b \times b$ minor of the vectors $(\bar{\ell}_{\omega'}, t_{\omega'} \bar{\ell}_{\omega'})$ on the columns J_0 (first copy of W) and J_1 (second copy). By the Laplace expansion along the column split (J_0, J_1) , the monomial $\prod_{\omega' \in B_1} t_{\omega'}$ comes only from the term that puts the rows B_1 on J_1 . Its coefficient is $\pm \det M_{B_0} \det M_{B_1} \neq 0$, and it does not involve the level- i variables. So the minor is nonzero and the b vectors are independent modulo $U \oplus U$.

The total rank is $2a + b = |S|$. □

Corollary 3.17 (Very large fields). *If $q - 1 \geq 2^\lambda n D \binom{L}{D}$ (with $D = D_n$, $L = L_n$), then with probability at least $1 - 2^{-\lambda}$ over uniform gates, C_n is MDS: $\Delta(C_n) = 1 - (D - 1)/L$.*

Proof. Parametrise each pair as $(t^0, t^1) = (a, a + s)$ with $(a, s) \in \mathbb{F} \times \mathbb{F}^\times$. This is a bijection onto distinct pairs, and an invertible linear change of variables.

For $|S| = D$, Theorem 3.16 makes $\det M_S$ a nonzero polynomial. Each entry of M_S is a product of at most n structural points, so $\deg \det M_S \leq nD$. Schwartz–Zippel over $\mathbb{F} \times \mathbb{F}^\times$ gives $\Pr[\det M_S = 0] \leq nD/(q - 1)$. Take a union bound over the $\binom{L}{D}$ sets. □

Remark 3.18. Corollary 3.17 needs $\log_2 q \gtrsim L H_2(R)$, so it is structural rather than practical. It shows that the level count in Theorem 3.14 is not forced by the construction. Over Goldilocks with random gates, every D -subset was checked to be nonsingular for $(n, k) \in \{(2, 1), (2, 2), (2, 3), (3, 1), (3, 2)\}$ (about $1.06 \cdot 10^7$ subsets).

A per-set bound cannot close the gap: for $|S| = D + 1$ the measured $\Pr[\text{rank } M_S < D]$ is about $1.5/q$, not about q^{-2} , because a single coincidence of gate points makes many sets degenerate at once.

3.3 Checked setup

The losses of Theorem 3.14 at the first levels come from short codes, where a union bound is expensive. These levels are small enough to be verified once, at setup.

Theorem 3.19 (Distance with a checked setup). *Fix $i_0 \geq 0$. Sample the gates of levels $\leq i_0$ (variables $x_n, \dots, x_{n-i_0+1}$) and resample them until C_{i_0} is MDS. Sample the gates of levels $> i_0$ independently and uniformly on $\mathcal{G}^*(\mathbb{F})$. Choose $g_{i_0+1}, \dots, g_n \geq 0$, put $Z_{i_0} = D_{i_0}$ and $Z_i = 2Z_{i-1} + g_i$, and assume $Z_i \leq L_i$. With probability at least*

$$1 - \sum_{i=i_0+1}^n \binom{L_i}{Z_i} \frac{(1 + 1/q)^{L_{i-1}}}{q - 1} q^{-g_i},$$

every nonzero codeword of C_n has fewer than Z_n zeros, so $\Delta(C_n) \geq 1 - (Z_n - 1)/L_n$.

Proof. C_{i_0} is MDS, so every nonzero codeword of C_{i_0} has at most $D_{i_0} - 1$ zeros: $V_S = 0$ whenever $|S| \geq Z_{i_0}$. The acceptance test depends only on the gates of levels $\leq i_0$, so the gates of levels $> i_0$ are still independent and uniform after conditioning on acceptance. Run the induction of Theorem 3.14 from $i = i_0 + 1$. $\square$

Remark 3.20 (Cost of the check). C_1 is MDS iff the 2^{k+1} points of the level-1 gates are pairwise distinct. C_2 is MDS iff every 4-subset of its 2^{k+2} positions gives a nonsingular 4×4 matrix with rows $(1, y_1, y_2, y_1 y_2)$. By Theorem 3.16 and Schwartz–Zippel, a uniform family is rejected with probability at most $\binom{L_{i_0}}{D_{i_0}} i_0 D_{i_0} / (q - 1)$, which is negligible for $i_0 \leq 2$ at $q \approx 2^{64}$. The check is done once; the verifier only recomputes the gates from the public seed and the accepted counter.

gate field	copies	$i_0 = 0$	$i_0 = 1$	$i_0 = 2$	$i_0 = 3$
M31 (2^{31})	64	0.417	0.527	0.617	0.683
Goldilocks (2^{64})	16	0.579	0.669	0.742	0.768
Goldilocks (2^{64})	32	0.729	0.789	0.822	0.854
Goldilocks (2^{64})	64	0.811	0.865	0.888	0.905

Table 4: Certified $\Delta(C_n)$ from Theorem 3.19, $n = 20$, setup failure $\leq 2^{-128}$. The column $i_0 = 0$ is Theorem 3.14.

3.4 Zero structure

For $F \in \text{ML}(x_1, \dots, x_n)$ and a prefix $c = (\beta, \theta_n, \dots, \theta_{n-i+1}) \in \Omega_i$, let $F_c \in \text{ML}(x_1, \dots, x_{n-i})$ be F with x_l replaced by $r_l(c)$ for $l > n - i$ (Proposition 3.3). The prefix c is *killed* if $F_c = 0$, and *maximally killed* if moreover its parent is not killed.

$\log_2 q$ (gate field)	copies	Δ	$1 - R$	queries
64	16	0.742	0.938	210
64	32	0.822	0.969	180
128	8	0.757	0.875	201
128	16	0.863	0.938	169
256	8	0.836	0.875	176
256	16	0.913	0.938	157

Table 5: Gates in a larger field (e.g. an extension of the coefficient field), $i_0 = 2$, $n = 20$, $\lambda = 128$. Queries for 2^{-128} soundness by Theorem 6.4 with $\delta = 0.95 \delta_{\max}$. A larger gate field reduces the number of copies; the number of queries is bounded below by the radius $\delta < \Delta/2$ of Lemma 6.2.

Lemma 3.21 (Zero decomposition). *Let $F \neq 0$ and let $\kappa_i(F)$ be the number of maximally killed prefixes in Ω_i . The zero set of $\text{Enc}_n(F)$ is the disjoint union of the subtrees below the maximally killed prefixes, and*

$$\#\{\omega : \text{Enc}_n(F)(\omega) = 0\} = \sum_{i=1}^n \kappa_i(F) 2^{n-i}.$$

Moreover, if $p \in \Omega_{i-1}$ is not killed, at most one child of p is killed. For fixed F and fixed gates of the levels above p , a given child of p is killed with probability at most $1/q$ over the gate at p .

Proof. By Proposition 3.3, $\text{Enc}_n(F)(\omega) = F_\omega$, so the zeros are the killed leaves. On the path from the root to a zero leaf, $F_{(\beta)} = F \neq 0$ and the leaf is killed, so there is a first killed prefix; it is maximally killed. Every prefix below a killed prefix is killed, since a restriction of 0 is 0. Two maximally killed prefixes are not comparable, so their subtrees are disjoint, and a prefix in Ω_i has 2^{n-i} leaves.

Write $F_p = A + xB$ with $x = x_{n-i+1}$. The children are $A + t^0(p)B$ and $A + t^1(p)B$. If both vanish, then $(t^0(p) - t^1(p))B = 0$, so $B = 0$, $A = 0$ and $F_p = 0$. A child is killed iff $A = -tB$ for its point t : this is impossible if $B = 0$ (then $A \neq 0$), and otherwise holds for at most one value of t . $\square$

Proposition 3.22 (Coincidences cost $R2^{-j}$). *Let $1 \leq j \leq n$ and let $p \neq p' \in \Omega_{j-1}$ lie in different copies ($\beta \neq \beta'$). If two structural points of their gates coincide, $t^b(p) = t^{b'}(p') = s$, then*

$$\Delta(C_n) \leq 1 - R(1 + 2^{-j}) + \frac{1}{L_n}.$$

Proof. Write $p = (\beta, \theta_n, \dots, \theta_{n-j+2})$. For $1 \leq i \leq j-1$ let c_i be the structural point of the sibling, off the path of p , at level i : $c_i = t_{n-i+1}^{1-\theta_{n-i+1}}(\beta, \theta_n, \dots, \theta_{n-i+2})$. Put

$$F = \prod_{i=1}^{j-1} (x_{n-i+1} - c_i) \cdot (x_{n-j+1} - s) \cdot K(x_1, \dots, x_{n-j}),$$

which is multilinear. It vanishes on:

- the $j-1$ sibling subtrees off the path of p , of sizes $2^{n-1}, \dots, 2^{n-j+1}$, which sum to $2^n - 2^{n-j+1}$;
- the subtrees below (p, b) and (p', b') , of size 2^{n-j} each. They are disjoint from the previous ones ((p, b) is on the path of p , and p' is in another copy).

This gives 2^n zeros. The factor K ranges over a space of dimension 2^{n-j} , so it can be chosen nonzero and vanishing at $2^{n-j} - 1$ further positions. Then $F \neq 0$ has at least $2^n + 2^{n-j} - 1 = D_n - 1 + 2^{n-j}$ zeros, a relative weight of at most $1 - R - R 2^{-j} + 1/L_n$. $\square$

Remark 3.23 (Consequences for the level count). Each pair of points of distinct level- j nodes coincides with probability $1/q$. So a theorem with failure probability $2^{-\lambda} < 1/q$ must allow a loss of order $(\lambda/\log_2 q) R 2^{-i_0}$ after a checked setup at level i_0 . Coincidences are not the only source of excess zeros: for $n = 2$, four positions in different level-2 nodes whose points lie on a line $x_2 = a + bx_1$ give the nonzero $F = x_2 - a - bx_1$ with $4 = D_2$ zeros, although all points are distinct. A proof that removes the level count must therefore control algebraic relations among many gate points at once.

Open problem 3.24 (Removing the level count). Theorems 3.14 and 3.19 lose about $H_2(2^{-k})/\log_2 q$ per level. Prove a bound

$$\Delta(C_n) \geq 1 - R - c_1 \frac{H_2(R)}{\log_2 q} - c_2 \frac{\lambda}{\log_2 q} R 2^{-i_0} - o(1)$$

with c_1, c_2 independent of n , or show that a loss growing with n is unavoidable.

4 Folding

Definition 4.1 (Fold). For $1 \leq j \leq n$ and $z \in \mathbb{K}$, define $\text{fold}_{j,z} : \mathbb{K}^{\Omega_{n-j+1}} \rightarrow \mathbb{K}^{\Omega_{n-j}}$ by

$$(\text{fold}_{j,z} w)(\omega') = w(\omega', 0) + \lambda_{j,\omega'}(z)(w(\omega', 1) - w(\omega', 0)), \quad \lambda_{j,\omega'}(z) = \frac{z - t_j^0(\omega')}{t_j^1(\omega') - t_j^0(\omega')}.$$

By Lemma 2.4, this equals the gate rule of $A_j(\omega')$ at $x_j = z$:

$$(\text{fold}_{j,z} w)(\omega') = W_L(z) \frac{d}{\delta} w(\omega', 1) - W_R(z) \frac{c}{\delta} w(\omega', 0).$$

Lemma 4.2 (Compatibility). For $F \in \text{ML}(x_j, \dots, x_n)$,

$$\text{fold}_{j,z}(\text{Enc}_{n-j+1}(F)) = \text{Enc}_{n-j}(F|_{x_j=z}).$$

Consequently, folding $\text{Enc}_n(F)$ with $z_1, \dots, z_n$ gives the constant word $F(z_1, \dots, z_n)$.

Proof. With $F = u + x_j v$, the block at ω' is $(e_u + t^0 e_v, e_u + t^1 e_v)$, where $e_u = \text{Enc}(u)(\omega')$ and $e_v = \text{Enc}(v)(\omega')$. Its fold is $e_u + z e_v$, and $\text{Enc}(u) + z \text{Enc}(v) = \text{Enc}(u + z v)$ by linearity. $\square$

Lemma 4.3 (Contraction). $d_{\text{pos}}(\text{fold}_{j,z} w, \text{fold}_{j,z} w') \leq d_{\text{blk}}(w, w')$.

Proof. $(\text{fold} w)(\omega')$ depends only on the block of w at ω' . $\square$

Lemma 4.4 (Line form). For $w \in \mathbb{K}^{\Omega_{n-j+1}}$ define $A, B \in \mathbb{K}^{\Omega_{n-j}}$ by gate inversion:

$$B(\omega') = \frac{w(\omega', 1) - w(\omega', 0)}{t_j^1(\omega') - t_j^0(\omega')}, \quad A(\omega') = w(\omega', 0) - t_j^0(\omega') B(\omega').$$

Then $\text{fold}_{j,z} w = A + zB$ for all z . The map $w \mapsto (A, B)$ is bijective. For $a, b \in C_{n-j}$, the word w agrees with the codeword of C_{n-j+1} built from (a, b) on exactly the blocks ω' where $(A, B)(\omega') = (a, b)(\omega')$.

Proof. Direct from Definition 3.2 and $t^0 \neq t^1$. $\square$

Lemma 4.5 (Line proximity). *Let $C \subseteq \mathbb{K}^N$ be linear with $\Delta(C) \geq \Delta$, let $0 \leq \delta < \Delta/3$, and $A, B \in \mathbb{K}^N$. Let $\mathcal{Z} = \{z \in \mathbb{K} : d_{\text{pos}}(A + zB, C) \leq \delta\}$. If $|\mathcal{Z}| = M \geq 2$, there exist $a, b \in C$ with*

$$|\{p : (A(p), B(p)) \neq (a(p), b(p))\}| \leq \frac{M}{M-1} \delta N.$$

Proof. Take $z_1 \neq z_2$ in $\mathcal{Z}$, with codewords c_1, c_2 agreeing with $A + z_i B$ on sets S_1, S_2 of size $\geq (1 - \delta)N$. Put $b = (c_1 - c_2)/(z_1 - z_2)$ and $a = c_1 - z_1 b$, both in C . On $S_1 \cap S_2$ we have $A = a$ and $B = b$.

For any $z_3 \in \mathcal{Z}$ with codeword c_3 and agreement set S_3 , the codewords c_3 and $a + z_3 b$ agree on $S_1 \cap S_2 \cap S_3$. This set has size $\geq (1 - 3\delta)N > (1 - \Delta)N$, so $c_3 = a + z_3 b$. Hence every $z \in \mathcal{Z}$ has agreement set $S_z \subseteq \{p : A(p) + zB(p) = a(p) + zb(p)\}$.

Let $E = \{p : (A, B)(p) \neq (a, b)(p)\}$. For $p \in E$ the affine function $z \mapsto (A - a)(p) + z(B - b)(p)$ is nonzero, so $p \in S_z$ for at most one z . Then $\sum_{z \in \mathcal{Z}} |E \cap S_z| \leq |E|$. Combined with $|E \setminus S_z| \leq \delta N$, this gives $M|E| - |E| \leq M\delta N$. $\square$

Corollary 4.6 (Folding preserves distance). *Let $\Delta_{n-j} = \Delta(C_{n-j})$, $0 \leq \delta < \Delta_{n-j}/3$ and $\eta > 0$. If $d_{\text{blk}}(w, C_{n-j+1} \otimes \mathbb{K}) \geq (1 + \eta)\delta$, then*

$$|\{z \in \mathbb{K} : d_{\text{pos}}(\text{fold}_{j,z} w, C_{n-j} \otimes \mathbb{K}) \leq \delta\}| \leq 1 + \eta^{-1}.$$

Proof. Write $\text{fold}_{j,z} w = A + zB$ (Lemma 4.4) and let M be the number of such z . If $M \geq 2$, Lemma 4.5 and Lemma 4.4 give $d_{\text{blk}}(w, C_{n-j+1} \otimes \mathbb{K}) \leq \frac{M}{M-1} \delta$. Hence $1 + \eta \leq M/(M-1)$, i.e. $M \leq 1 + \eta^{-1}$. Lemma 3.6 transfers Δ to $\mathbb{K}$. $\square$

5 The commitment scheme

Parameters. n (variables), k (redundancy), s (queries), $\mathbb{F} \subseteq \mathbb{K}$, a hash function H , and a public seed. The gate family $\mathcal{A}$ is derived from the seed with gates uniform on $\mathcal{G}^*(\mathbb{F})$. Merkle leaves are blocks: the leaf of a layer at ω' is the pair $(w(\omega', 0), w(\omega', 1))$.

5.1 Protocol

1. **Commit.** The prover computes $E_0 = \text{Enc}_n(F) \in \mathbb{F}^{\Omega_n}$ and publishes its Merkle root R_0 .
2. **Fold.** For $j = 1, \dots, n$: the verifier sends $z_j \in \mathbb{K}$ uniform (in the non-interactive version $z_j = H(R_0, \dots, R_{j-1})$). The prover computes $E_j = \text{fold}_{j,z_j}(E_{j-1}) \in \mathbb{K}^{\Omega_{n-j}}$. For $j < n$ it publishes the root R_j . For $j = n$ it sends $E_n \in \mathbb{K}^{2^k}$ in full.
3. **Output.** The verifier checks that E_n is constant and outputs $y = E_n(\beta)$, the claimed value of $F(z_1, \dots, z_n)$.
4. **Query.** Repeat s times: sample $\omega = (\beta, \theta_n, \dots, \theta_1) \in \Omega_n$ uniformly. For $j = 1, \dots, n$, open the block of E_{j-1} at $\omega^{(j)}$ and the entry $E_j(\omega^{(j)})$; the latter lies in the block of E_j at $\omega^{(j+1)}$, opened in the next step, or in E_n . Check

$$E_j(\omega^{(j)}) = W_L(z_j) \frac{d}{\delta} E_{j-1}(\omega^{(j)}, 1) - W_R(z_j) \frac{c}{\delta} E_{j-1}(\omega^{(j)}, 0)$$

with $A_j(\omega^{(j)}) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, together with all Merkle paths.

Per query: n block openings, Merkle paths of lengths $n + k - j$, and $O(n)$ field operations.
Prover: $(n + 1)2^{n+k}$ operations to encode and 2^{n+k} to fold, plus hashing.

6 Security

Theorem 6.1 (Completeness). *An honest prover is always accepted, and $y = F(z_1, \dots, z_n)$.*

Proof. Lemma 4.2 and Lemma 2.4. $\square$

For words x, x' on the same layer write $\text{Agr}(x, x') = \{\omega : x(\omega) = x'(\omega)\}$, and for $w, c \in \mathbb{K}^{\Omega_i}$ ($i \geq 1$) write $\text{BAgr}(w, c) \subseteq \Omega_{i-1}$ for the set of blocks on which w and c agree in both entries.

Lemma 6.2 (Folding preserves agreement sets). *Let $1 \leq j \leq n$ and $C = C_{n-j} \otimes \mathbb{K}$ with $\Delta(C) \geq \Delta$. Let $\varepsilon > 0$ and*

$$0 \leq \delta < \delta_{\max}(\Delta) := \min\left(1 - (1 - \Delta)^{1/3}, \frac{\Delta}{2}\right).$$

Put

$$\gamma = \frac{(1 - \delta)^3 - (1 - \Delta)}{\Delta}, \quad M = \left\lfloor \frac{\delta}{\Delta - 2\delta} \right\rfloor + 2, \quad N_1 = \left\lceil \max\left(\frac{12}{\Delta\gamma}, \frac{2M}{\gamma} + 2\right) \right\rceil.$$

For every $w \in \mathbb{K}^{\Omega_{n-j+1}}$ there is a set $B(w) \subseteq \mathbb{K}$ with $|B(w)| < N_1 + 1/\varepsilon$ such that, for every $z \notin B(w)$ and every $c' \in C$ with $d_{\text{pos}}(\text{fold}_{j,z}w, c') \leq \delta$, there is $c \in C_{n-j+1} \otimes \mathbb{K}$ with

$$\text{fold}_{j,z}(c) = c' \quad \text{and} \quad |\text{Agr}(\text{fold}_{j,z}w, c') \setminus \text{BAgr}(w, c)| \leq \varepsilon L_{n-j}.$$

Proof. Write $L = L_{n-j}$ and $\text{fold}_{j,z}w = A + zB$ (Lemma 4.4). Let $\mathcal{Z} = \{z : d_{\text{pos}}(A + zB, C) \leq \delta\}$. Since $2\delta < \Delta$, each $z \in \mathcal{Z}$ has a unique codeword c_z within distance δ ; let S_z be its agreement set, $|S_z| \geq (1 - \delta)L$. Consider the points $P_z = (z, c_z)$ of the affine space $\mathbb{K} \times C$.

Step 1 (good triples are collinear). Let $z_1, z_2, z_3 \in \mathcal{Z}$ be distinct with $|S_{z_1} \cap S_{z_2} \cap S_{z_3}| > (1 - \Delta)L$. Put $b = (c_{z_1} - c_{z_2})/(z_1 - z_2)$ and $a = c_{z_1} - z_1b$. On $S_{z_1} \cap S_{z_2}$ we have $(A, B) = (a, b)$. On the triple intersection, $c_{z_3} = A + z_3B = a + z_3b$. Two codewords agreeing on more than $(1 - \Delta)L$ positions are equal, so $c_{z_3} = a + z_3b$: the three points lie on one line.

Step 2 (many good triples). Let $N = |\mathcal{Z}| \geq 3$ and $Z_p = \{z \in \mathcal{Z} : p \in S_z\}$. Then $\sum_p |Z_p| \geq N(1 - \delta)L$, and the sum of $|S_{z_1} \cap S_{z_2} \cap S_{z_3}|$ over ordered triples of distinct elements equals $\sum_p |Z_p|(|Z_p| - 1)(|Z_p| - 2)$. Since $x(x - 1)(x - 2) \geq \max(0, x - 2)^3$ on integers and the right side is convex, this sum is at least $L(N(1 - \delta) - 2)^3$. Dividing by N^3 , the average triple intersection is at least $L(1 - \delta - 2/N)^3$. Each triple intersection is at most L , and at most $(1 - \Delta)L$ for a bad triple. So the fraction of good triples is at least

$$\gamma_N = \frac{(1 - \delta - 2/N)^3 - (1 - \Delta)}{\Delta} \geq \gamma - \frac{6}{N\Delta}.$$

Step 3 (a heavy line). Distinct points of $\mathbb{K} \times C$ span a unique line. If line ℓ carries m_ℓ of the points, ordered collinear triples number $\sum_\ell m_\ell(m_\ell - 1)(m_\ell - 2) \leq (\max_\ell m_\ell) N(N - 1)$. Hence some line carries at least $\gamma_N(N - 2)$ points. For $N \geq N_1$ this is at least M .

Step 4 (agreement along a line). Let a line carry $m \geq 2$ points, i.e. $c_z = a + zb$ for m values z . Put $E = \{p : (A, B)(p) \neq (a, b)(p)\}$. As in Lemma 4.5, each $p \in E$ lies in at most one S_z , while $|E \setminus S_z| \leq \delta L$. Hence $|E| \leq \frac{m}{m-1} \delta L$.

Step 5 (absorption). Let a line ℓ with pair (a, b) carry $m \geq M$ points, and let E be as in Step 4, so $|E| \leq \frac{M}{M-1} \delta L$. For any $z \in \mathcal{Z}$, the codewords c_z and $a + zb$ both equal $A + zB$ on $S_z \setminus E$, a

set of size at least $(1 - \delta - \frac{M}{M-1}\delta)L$. Since $M - 1 > \frac{\delta}{\Delta - 2\delta}$, we have $\delta(2 + \frac{1}{M-1}) < \Delta$, so this size exceeds $(1 - \Delta)L$ and $c_z = a + zb$. Hence *every* point P_z , $z \in \mathcal{Z}$, lies on ℓ .

Step 6 (the exceptional set).

- If $N < N_1$, put $B(w) = \mathcal{Z}$.
- Otherwise Step 3 gives a line ℓ carrying at least M points, with pair (a, b) and set E , and Step 5 puts all of $\mathcal{Z}$ on ℓ .
- On ℓ we have $S_z = E^c \cup (E \cap S_z)$ and $\sum_{z \in \ell} |E \cap S_z| \leq |E| \leq L$. So fewer than $1/\varepsilon$ values $z \in \ell$ have $|E \cap S_z| > \varepsilon L$.
- Put $B(w) = \{z \in \mathcal{Z} : |E \cap S_z| > \varepsilon L\}$.

For $z \notin B(w)$ with a codeword c' within δ , we have $z \in \ell$ and $c' = c_z = a + zb$. Let c be the word of $C_{n-j+1} \otimes \mathbb{K}$ built from (a, b) . Then $\text{fold}_{j,z} c = c'$, and $\text{BAgr}(w, c) = E^c$ by Lemma 4.4. $\square$

Remark 6.3. For $\Delta < 3 - \sqrt{5} \approx 0.76$ the binding term is the triple radius $1 - (1 - \Delta)^{1/3}$, which exceeds $\Delta/3$. Example: for $\Delta = 0.508$ the radius grows from 0.169 to 0.21.

Fix a family with $\Delta(C_i) \geq \Delta^*$ for all i , a threshold $0 < \delta < \delta_{\max}(\Delta^*)$ (Lemma 6.2) and $\varepsilon > 0$. Say the commitment R_0 binds $F^* \in \text{ML}(x_1, \dots, x_n)$ (coefficients in $\mathbb{K}$) if $d_{\text{blk}}(E_0, \text{Enc}_n(F^*)) < \delta$. Such an F^* is unique, because $2\delta < \Delta^*$.

Theorem 6.4 (Soundness). *For any prover, the probability that the verifier accepts while either R_0 binds no polynomial, or R_0 binds F^* and $y \neq F^*(z_1, \dots, z_n)$, is at most*

$$\frac{n(N_1 + 1/\varepsilon)}{|\mathbb{K}|} + (1 - \delta + n\varepsilon)^s$$

in the IOP model (Merkle binding assumed).

Proof. Write $f_j = \text{fold}_{j,z_j}(E_{j-1})$; the check of step j on a query passes iff $E_j(\omega^{(j)}) = f_j(\omega^{(j)})$.

Good challenges. E_{j-1} is fixed before z_j is drawn. Let G be the event $z_j \notin B(E_{j-1})$ for all j (Lemma 6.2). Then $\Pr[\neg G] \leq n(N_1 + 1/\varepsilon)/|\mathbb{K}|$. Assume G , and that the verifier checked that E_n is the constant y .

Descent. Put $c'_n = E_n \in C_0 \otimes \mathbb{K}$. For $j = n, n-1, \dots, 1$:

- if $d_{\text{pos}}(f_j, c'_j) > \delta$, stop and set $J = j$;
- otherwise, Lemma 6.2 gives c'_{j-1} with $\text{fold}_{j,z_j} c'_{j-1} = c'_j$ and a set $X_j = \text{Agr}(f_j, c'_j) \setminus \text{BAgr}(E_{j-1}, c'_{j-1})$ with $|X_j| \leq \varepsilon L_{n-j}$.

If the descent never stops, set $J = 0$. The codewords c'_j and sets X_j depend only on the transcript before the query phase.

One query. Let ω be a query and $p_j = \omega^{(j)}$. Suppose it passes every check and $p_j \notin X_j$ for all $j > J$. We show by descending induction that $E_j(p_j) = c'_j(p_j)$ for $j \geq J$.

- For $j = n$ this is the definition of c'_n .
- Assume it for $j > J$. The check at step j gives $f_j(p_j) = E_j(p_j) = c'_j(p_j)$, so $p_j \in \text{Agr}(f_j, c'_j)$. Since $p_j \notin X_j$, the block of E_{j-1} at p_j agrees with c'_{j-1} in both entries. In particular $E_{j-1}(p_{j-1}) = c'_{j-1}(p_{j-1})$.

Consequently:

- if $J \geq 1$, then $p_J \in \text{Agr}(f_J, c'_J)$, a set of relative size $< 1 - \delta$;
- if $J = 0$, the block of E_0 at p_1 lies in $\text{BAgr}(E_0, c'_0)$, a set of relative size $1 - d_{\text{blk}}(E_0, c'_0)$.

Since each p_j is uniform on its layer,

$$\Pr[\text{query passes}] \leq n\varepsilon + \max(1 - \delta, 1 - d_{\text{blk}}(E_0, c'_0)).$$

The bad cases. Folding c'_0 through all rounds gives c'_n , the constant y . Write $c'_0 = \text{Enc}_n(\tilde{F})$; Lemma 4.2 gives $y = \tilde{F}(z)$.

- If R_0 binds no polynomial, then $d_{\text{blk}}(E_0, c'_0) \geq \delta$.
- If R_0 binds F^* and $y \neq F^*(z)$, then $\tilde{F} \neq F^*$. Distinct codewords are at block distance $\geq \Delta^*$, so $d_{\text{blk}}(E_0, c'_0) \geq \Delta^* - \delta > \delta$.

In both cases a query passes with probability $\leq 1 - \delta + n\varepsilon$. The s query positions are independent of everything fixed above. $\square$

Remark 6.5 (Why this bound has no factor n). The argument follows the path of a query from the constant y down to T_0 . Lemma 6.2 guarantees that whenever a check passes on a position, the whole pair below it, including the sibling not on the path, agrees with one fixed valid table. A layer-by-layer distance count cannot see this, and loses a factor n .

6.1 Committing only some layers

Folding is a public deterministic map, so a layer that is not committed can be recomputed by the verifier from the layer above it. This gives folding by 2^m per round with no new soundness loss.

Definition 6.6 (Commitment schedule). A *schedule* is a set $J \subseteq \{1, \dots, n-1\}$ of layers that are *not* committed. The protocol Π_J is the protocol of §5 with these changes.

- For $j \in J$ the prover sends no root R_j . The challenge z_{j+1} is sampled right after z_j , with no prover message in between (non-interactively: $z_{j+1} = H(\text{transcript}, z_j)$).
- In the query phase, for a committed layer $j_0 \notin J$ followed by the run $j_0 + 1, \dots, j_0 + m - 1 \in J$ and the next committed layer $j_0 + m \notin J$ (or $j_0 + m = n$), the verifier opens the whole group of 2^m entries of E_{j_0} below the prefix $\omega^{(j_0+m)}$, with one Merkle path. It recomputes the values of the skipped layers on this group with the gate rule, and checks the resulting value against the opened entry of E_{j_0+m} (or against E_n).

Merkle leaves of layer j_0 are groups of 2^m entries. $J = \emptyset$ is the protocol of §5; $J = \{1, 3, 5, \dots\}$ is folding by 4 per round.

Theorem 6.7 (Soundness with any schedule). *For every J , the soundness error of Π_J is at most the bound of Theorem 6.4:*

$$\frac{n(N_1 + 1/\varepsilon)}{|\mathbb{K}|} + (1 - \delta + n\varepsilon)^s.$$

Proof. Work in the IOP model of Theorem 6.4: a root is an oracle to the full table, so the grouping of Merkle leaves plays no role. Let P^* be any prover for Π_J . Build a prover P' for $\Pi_\emptyset$:

- P' runs P^* and forwards its messages and roots.
- For $j \in J$, after receiving z_j , P' defines $E_j := \text{fold}_{j,z_j}(E_{j-1})$, where E_{j-1} is the committed table, or the table defined in the same way if $j-1 \in J$. It commits to E_j honestly and sends the root R_j .
- In the query phase P' answers the openings of committed layers with the answers of P^* , and the openings of the layers $j \in J$ honestly from E_j .

Same challenges. In both protocols each z_j is uniform on $\mathbb{K}$ and independent of everything before it. Inserting the root R_j ($j \in J$) does not change this, because E_j and hence R_j are functions of messages already sent.

Same acceptance. Fix a query leaf ω and a run $j_0, \dots, j_0 + m$ as in the definition.

- The blocks of E_{j_0} that $\Pi_\emptyset$ opens along ω in this run are contained in the group that Π_J opens, and the Π_J verifier sees them under one Merkle path to R_{j_0} .
- For $j \in J$, the $\Pi_\emptyset$ check at layer j compares $E_j(\omega^{(j)})$ with the gate rule applied to a block of E_{j-1} . Since $E_j = \text{fold}_{j,z_j}(E_{j-1})$ entrywise, it passes.
- The $\Pi_\emptyset$ check at the committed layer $j_0 + m$ compares the opened entry of E_{j_0+m} with the gate rule applied to a block of E_{j_0+m-1} . That block is the recomputed value of the group, so this is exactly the Π_J check.

Hence the $\Pi_\emptyset$ verifier accepts against P' whenever the Π_J verifier accepts against P^* , on the same randomness, with the same output y and the same root R_0 . Theorem 6.4 bounds the probability of the bad event for P' . $\square$

Remark 6.8 (Costs). Committing every m -th layer gives $\lceil n/m \rceil$ Merkle trees, and per query $\lceil n/m \rceil$ paths, each with a leaf of 2^m entries. The verifier work for the gate rule is unchanged, about $2^m - 1$ gate evaluations per committed layer. The same argument covers sending the last r layers in full: the verifier folds E_{n-r} itself and checks that the result is constant. The query count s and the challenge field are unchanged.

Remark 6.9 (Non-interactive version). With $z_j = H(\text{transcript up to } R_{j-1})$ and query positions hashed from the full transcript, a prover making Q hash queries gains at most a factor Q on the first term, plus the Merkle collision term.

Remark 6.10 (Concrete parameters, proven). Take $\mathbb{F} = \mathbb{F}_{2^{31}-1}$, $n = 20$, $k = 6$ (64 copies, $\Delta^* \geq 0.508$), $\mathbb{K}$ of degree 5 over $\mathbb{F}$, $\delta = 0.20$ (so $\gamma \approx 0.0394$, $M = 3$, $N_1 = 600$) and $\varepsilon = 2^{-20}$. Then $n(N_1 + 1/\varepsilon)/|\mathbb{K}| \leq 2^{-130}$, and $s = 398$ queries give $(1 - \delta + n\varepsilon)^s < 2^{-128}$.

Remark 6.11 (Any field). Nothing above uses structure of $\mathbb{F}$ beyond its size: no subgroup and no special element. In particular every result holds in characteristic 2: the proofs use only $t^0 \neq t^1$, division by $t^1 - t^0$, and Schwartz–Zippel. In a binary field, additions are XOR and Lemma 2.5 reads $\sum_{x \in \{0,1\}} F = cF_L + dF_R$. If $\mathbb{F}$ is too small for Theorem 3.10 (roughly $q < 2^{30}$ at $n = 20$), sample the gates in an extension $\mathbb{F}' \supseteq \mathbb{F}$. The table E_0 then lives in $\mathbb{F}'$, and every statement holds with $q = |\mathbb{F}'|$. The challenge field $\mathbb{K}$ only needs $n(N_1 + 1/\varepsilon)/|\mathbb{K}|$ to be negligible.

Open problem 6.12 (Up to $\Delta^*/2$). Remove the triple-radius condition in Lemma 6.2, so that $\delta < \Delta^*/2$ suffices. With $\Delta^* = 0.508$ this gives about 303 queries.

7 Distance in practice

Theorem 3.14 loses about $n H_2(2^{-k})/\log_2 q$ in total. At realistic field sizes the experiments below found no codeword below the Singleton limit; whether a loss growing with n is intrinsic is open (Problem 3.24).

Experiment. Over $\mathbb{F}_{2^{31}-1}$, with per-node random gates:

- **Rank tests.** Test sets of 2^n positions for linear dependence of their columns, both uniformly random sets and unions of whole subtrees, where the tree structure would show a weakness.
- **Zero search.** Choose $2^n - 1$ positions, take the nonzero polynomial vanishing there, and count all its zeros. More than $2^n - 1$ zeros would beat the Singleton limit.

n	copies	length	2^n	singular (random)	singular (subtrees)	max zeros	Δ found
6	2	128	64	0/300	0/300	63	0.508
6	4	256	64	0/300	0/300	63	0.754
6	8	512	64	0/300	0/300	63	0.877
8	2	512	256	0/120	0/120	255	0.502
8	4	1024	256	0/120	0/120	255	0.751
8	8	2048	256	0/120	0/120	255	0.876

Table 6: Every test meets the Singleton limit exactly. Control with per-layer gates ($n = 6$): 225 to 300 singular sets out of 300, and codewords of relative weight 2^{-6} , as Proposition 3.7 predicts.

setting	copies	Δ	queries	size of E_0	status
gates in Goldilocks	64	0.811	183	$6.7 \cdot 10^7$	proven (Thm. 3.14)
gates in Goldilocks	16	0.579	327	$1.7 \cdot 10^7$	proven (Thm. 3.14)
gates in Goldilocks ³	8	0.812	183	$8.4 \cdot 10^6$ in $\mathbb{F}_{p^3}$	proven (Thm. 3.14)
gates in M31	64	0.508	398	$6.7 \cdot 10^7$	proven (Thm. 3.10)
any field, $q \geq 2^{30}$	8	≈ 0.86	≈ 163	$8.4 \cdot 10^6$	if the level count is removed

Table 7: Parameters for $n = 20$ at 2^{-128} soundness (Theorem 6.4).

8 Further algebraic facts

All statements below were checked numerically over $\mathbb{F}_{2^{31}-1}$ (Rust test suite).

1. **Kronecker form.** Let the bottom table T_0 come from per-layer gates $A_1, \dots, A_n$, where layer i splits x_{n-i+1} and children are ordered (F_L, F_R) . With the coefficient vector α indexed so that bit $n - 1$ corresponds to x_n ,

$$\alpha = (A_1 \otimes A_2 \otimes \dots \otimes A_n) T_0$$

(A_1 acting on the most significant index bit), and $\det(\bigotimes_i A_i) = \prod_i \det(A_i)^{2^{n-1}}$.

2. **Sum row.** With $r(A) = [2a + c, 2b + d]$, $\sum_{x \in \{0,1\}^n} F(x) = (r(A_1) \otimes \dots \otimes r(A_n)) T_0$.

3. **Representation.** $\rho : \text{GL}_2(\mathbb{F})^n \rightarrow \text{GL}(\mathbb{F}^{2^n})$, $\rho(A_1, \dots, A_n) = \bigotimes_i A_i$, is a homomorphism with

$$\ker \rho = \left\{ (\lambda_1 I, \dots, \lambda_n I) : \prod_i \lambda_i = 1 \right\}.$$

The folding map $\alpha \mapsto T_0$ is $\rho(A)^{-1} = \rho(A^{-1})$. The assignment $A \mapsto \rho(A)^{-1}$ is an anti-homomorphism. The homomorphism to use is $A \mapsto \rho(A^{-\top})$.

4. **Identity gate.** For $A_i = I$, T_0 is the coefficient vector with index bits reversed relative to the splitting order.